FROM THE

EDITOR-IN-CHIEF

Health checkups are routine these days and progressively new tools are being invented to give you a more accurate diagnosis. Now, in the age of Artificial Intelligence, we are seeing a quantum leap in medical diagnostic technologies. In other domains, AI tools have of late become playthings for the public. But those such as ChatGPT or the ones that deal with image and video are producing a form of virtual reality. In healthcare, on the other hand, AI-powered technologies are getting as real as possible, cutting close to the bone with predictions on your body that are as solidly reliable as possible. Are you in line to get cancer? How much of a risk are you at for a heart attack or a cerebral stroke? And most importantly, when? At what stage will you see an onset of metabolic disorder, with its spiral of disease-causing conditions like diabetes and blood pressure? How much time before your arteries start choking like a traffic jam in your city? Do you have a strong predisposition for glaucoma-induced blindness? Does any form of dementia lurk around the corner in your sunset years? What about arthritis? Will you be allergic to Substance X or Y in the future?



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This is the future that really matters to you: that of your body. Now the new machines are designed to pick up the slightest changes in your body. They make the most minute and accurate readings. Everything shows up: any tiny bit of chemical misalignment in your blood, every slight blip in the markers that your vital organs produce and then the power of AI kicks in. Your readings get matched against the enormous mass of metadata available on the human body. The results are based on science and the power of analytics. They greatly assist the doctor in giving you the right advice and prescription.

It is understandable to be apprehensive about getting to know what will probably happen in the future but actually, our attitude ought to be quite the opposite. If you are not allergic to the truth, what you will open yourself up to is an embrace of predictive healthcare. It's an outright revolution in medical science, and its whole purpose is not to weigh you down with dark thoughts of mortality but to get you going and move towards solutions in good time. For what these machines are reading out to you is not your fate. Rather, this arms you with the capacity to thwart a negative occurrence before it happens. Cancer, for instance, may have already started forming in your body. In ordinary circumstances, it would remain imperceptible and without symptoms for years, till it has crossed the danger threshold. What you get here is really early detection. Information, thus, enables you to make sensible interventions at the very preliminary stages of a life-threatening phenomenon or before the onset of any form of debility.

This corresponds to an attitudinal turn within modern medical science. Traditional medicine systems such as Ayurveda had a very strong leaning towards prevention: they were embedded within systems of living that were meant to produce wellness. Allopathic medicine, on the other hand, had concentrated on treatment and cure. The variety of ways in which modern life was producing virtual epidemics of lifestyle diseases was not its primary concern. As Dr Naresh Trehan, Chairman of Medanta-The Medicity, tells INDIA TODAY, that started changing only in recent decades. "But 'preventive healthcare' was very general and didn't really motivate people to take the steps they needed to stay out of hospitals," he says. "Now, we are talking of 'predictive healthcare'—where we use multiple factors such as environment, genetics and a few blood parameters to see if one is at a higher risk for a non-communicable disease and come up with a preventive plan specific to an individual."

What predictive health offers is the highest form of risk analysis. This is still a statistical probability, not a confirmed guarantee of disease one way or the other. What matters here is that, once you ascertain the risk, steps can be taken to mitigate it. For example, a smoker who is told that his or her bodily analysis predicts a high risk of lung cancer or a heart attack can then be impelled to stop smoking. He or she may not fully eliminate the risk of disease but can certainly delay its likely onset by 10-15 years. Likewise, someone with a high risk of cancer can go for more frequent screenings to catch it at an early stage rather than be faced with the gloom and doom scenario of Stage III. Genetic testing can determine cases of familial hyperlipidemia: a blissfully ignorant passage through life for such an individual would have ended with abnormally high cholesterol. But the timely introduction of statins can avert a future heart attack. Well-informed medical interventions, thus, can ameliorate or completely reverse disease.

Besides scanning the whole spectrum of developments in this exciting new field, Senior Associate Editor Sonali Acharjee puts herself under the scanner to get a first-hand feel for this week's cover story. Checking into the Gurugram facility of Nura, a Fujifilm-Dr Kutty's Healthcare collaboration, to take their women's health screening package, she went through two hours of the most exhaustive series of tests. Exhaustive, but not exhausting. It was a breeze, she tells us. "I love getting medical tests done, even if I do feel a bit like a human guinea pig. I recently got a diagnosis that can predict risks of neurological decline based on the kind of metals found in your brain (losing my ability to think is my greatest fear!). There is still a long way to go before medical science can completely prevent your body's destiny. But for the time being, the awareness of what might or will happen can still help to maintain good health and wellbeing." India has only 1.3 million allopathic doctors for a population that's touching 1.43 billion. That's well short of the minimum WHO-mandated doctor-population ratio of 1:1,000. For a country such as ours, prevention via prediction can have vastly salutary effects. The good news is that these advanced diagnostic centres are mushrooming across the country. So, go for one of these checkups. Don't forget the old adage: Health is Wealth.



(Aroon Purie)